# Ram Representations and Local Certificates for the Periodic Spin-One Haldane Gap

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**Abstract**

We audit the pinned manuscript *The periodic spin-one Haldane gap*, distinguishing its analytic transfer construction, finite thermal and variational certificates, and coupled-purity bootstrap. We prove a sharp inverse from physical Gibbs purity to the first spectral gap and retain the source's exact terminal defects. Seven rationally checked restarts give, conditional on the original transfer and finite premises, $\gamma_L > 2363/500000$ for every even $L \geq 147456$ and $\Delta_1 \geq 2363/500000$, improving the source's eventual $\log(20)/784$ bound. We also prove a physical-Gram rounding bound for Eisenstein coordinates, reducing the first thermal table's certified error allowance from $10^{-5}$ to $9/2500000$, conditional on its printed totals, together with block-sensitive trace and weighted-moment certificates. These are local refinements, not an independent proof of the Haldane conjecture. Private Ram passages supply explicit representation, metric, remainder, and moment-certificate principles; the underlying spectral, convexity, and approximation techniques are classical. Complete thermal-column and trial-contraction recomputation remains a separate dependency. Contribution assessment: C3/5, proved local purity and physical-metric refinements strengthen conditional certificates but do not independently verify the source theorem or establish historical novelty.

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# 1 Pinned problem and claim boundaries

The assigned catalogue row is M543, family 268, at commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The source directory is

`_sources/openai-math/preprints/The-periodic-spin-one-Haldane-gap-September-24-2026/`.

All ten mathematical `build/*.tex` files were read, including every substantive section and all three mathematical appendices. The catalogue's L5 applicability rating is not the contribution score in this report. The source's own assertion that it proves the even-periodic Haldane gap is attributed to that source; this report does not convert a reading audit or partial computation into independent verification.

## 1.1 Hilbert spaces, operators, and boundaries

For every integer $L \geq 4$, the physical space is $\mathcal{H}_L = (\mathbb{C}^3)^{\otimes L}$ with its standard tensor-product inner product. All finite-dimensional operator domains are the entire indicated space. In the weight basis $s \in \{-1, 0, 1\}$,

$$S^z|s\rangle = s|s\rangle, \qquad S^+|s\rangle = \sqrt{2 - s(s+1)}\,|s+1\rangle, \qquad S^- = (S^+)^*.$$

Vectors outside the three allowed weights are interpreted as zero; $S^x = (S^+ + S^-)/2$ and $S^y = (S^+ - S^-)/(2i)$. Thus $[S^\alpha, S^\beta] = i\sum_\eta \varepsilon_{\alpha\beta\eta} S^\eta$ and $\sum_\alpha (S^\alpha)^2 = 2I$. With coupling exactly one, set

$$h = \sum_{\alpha=x,y,z} S^\alpha \otimes S^\alpha, \qquad H_L = \sum_{j=0}^{L-1} h_{j,j+1}, \quad j + L \equiv j. \tag{1.1}$$

No open end, endpoint field, anisotropy, or biquadratic term is included. The spectrum of a bond is $\{-2, -1, 1\}$. Define $E_0(L)$ as the least energy, $E_1(L)$ as the next *distinct* energy, and

$$\gamma_L = E_1(L) - E_0(L), \qquad \Delta_1 = \liminf_{\substack{L\to\infty \\ L \text{ even}}} \gamma_L.$$

Multiplicity of the ground energy is not removed when defining thermal weights. The model is nonscalar: the all-zero and all-+1 product vectors have different energy expectations. Therefore the next distinct eigenvalue exists.

For a Cartesian proper rotation $g \in SO(3)$, only the seam changes:

$$H_n(g) = \sum_{j=0}^{n-2} h_{j,j+1} + \sum_\alpha S^\alpha_{n-1}(gS^\alpha g^{-1})_0, \quad Z_n(b,g) = \operatorname{Tr} e^{-b(H_n(g)+anI)}, \quad a = \frac{700741}{500000}. \tag{1.2}$$

Here $b > 0$, $n \geq 4$, and traces are unnormalized. Write $Z_n(b) = Z_n(b, I)$. The twists used numerically are $I$, $P = \operatorname{diag}(1, -1, -1)$, and the cycle $C : e_x \mapsto e_y \mapsto e_z \mapsto e_x$. After global rotation their angles in the weight basis are $0, \pi, 2\pi/3$. A seam entry from column $y$ to row $x$ has phase $e^{i\theta(y_0 - x_0)}$. Odd $n$ occurs in the finite moment filters, although the final physical gap statement is for even $L$.

The source proves its elementary comparison

$$-\frac{3n}{2} I \leq H_n(g) \leq nI. \tag{1.3}$$

Its lower bound uses the three-site estimate $h_{12} + h_{23} \geq -3I$ and sums overlapping triads, counting each bond twice. A triad meeting the seam is gauge-equivalent to an untwisted open triad. This is not a frustration-free decomposition. Simultaneous similarity of the spin matrices preserves spectra under its tensor powers, but generally does not preserve Euclidean norms; all norm estimates below use the self-adjoint realization.

### 1.2 What the source claims and what is established here

The source's main theorem asserts unique ground states and

$$\gamma_L > \frac{4}{105} \log \frac{80}{79} \quad (L \geq 60 \text{ even}), \qquad \gamma_L > \frac{\log 20}{784} \quad (L \geq 2304 \text{ even}). \tag{1.4}$$

The second estimate implies $\Delta_1 \geq \log(20)/784$ within that proof. Finitely many smaller rings have positive gaps above their full ground sectors, without an explicit uniform small-length constant or simplicity conclusion from this observation alone.

Our new statements concern: (i) abstract purity and its exact inverse; (ii) conditional propagation using the source's finite premises; (iii) physical-coordinate rounding and trace certification; and (iv) the correct finite positive measure for spatial moments. We do not prove infinite-volume uniqueness, a boundary-field gap, a topological index, a rigorous numerical value near the physically estimated gap, or a computational speedup. A larger certified lower bound is a consequence relative to the pinned manuscript, not a claim of a historically new inequality.

## 2 Source dependency audit

### 2.1 Transfer operator before moments

For each $b > 0$, source `transfer.tex`, Proposition `prop:transfer`, defines $\Omega_b$ as the disjoint union of time-ordered bond histories

$$x = ((t_1, \alpha_1), \ldots, (t_k, \alpha_k)), \quad 0 < t_1 < \cdots < t_k < b, \quad \alpha_j \in \{x, y, z\},$$

including the empty history. Lebesgue measure on each labeled simplex and unit mass at the empty history give $\mu(\Omega_b) = e^{3b}$. The Hilbert space is $\mathcal{K}_b = L^2(\Omega_b, \mu; \mathbb{C})$, not $\mathcal{H}_L$ and not a finite moment-coordinate space. In Cartesian coordinates $G_\alpha = iS^\alpha$ is real with norm one. The kernel $k_b(x, y)$ is the trace of the merged chronological product of $G$ matrices; nonempty time collisions may be set to zero on a null set. It obeys

$$k_b(x, y) = k_b(y, x) \in \mathbb{R}, \quad |k_b(x, y)| \leq 3, \quad X_b = e^{-ab} K_b, \quad \|K_b\|_{\mathrm{HS}}^2 \leq 9e^{6b}.$$

Symmetry exchanges the two histories, not the chronological order of a matrix product. Hence $X_b$ is bounded, compact and self-adjoint, on all of $\mathcal{K}_b$, but need not be positive.

For the proper signed permutation rotation group, source `eq:twisted-kernel` constructs real commuting unitaries $U_g$ using both the permutation of history labels and the product of their signs. The Hilbert–Schmidt product trace identity, not an unsupported diagonal-kernel trace formula, identifies a cycle integral. Absolute traced terms of the ordered physical expansion sum to at most $3^n e^{3nb}$. This bound justifies regrouping by bond, including the rotated seam. Consequently the source obtains

$$Z_n(b, g) = \mathrm{Tr}(X_b^n U_g) \quad (n \geq 4), \qquad Z_n(b) = \sum_i \lambda_i(b)^n, \quad \sum_i |\lambda_i(b)|^k < \infty \ (k \geq 2). \tag{2.1}$$

The last sums count multiplicities. The source supplies the construction; a proposed finite Ram compression cannot replace its measure, seam identity, or trace-class justification.

### 2.2 Sector multiplicities and filters

The $D_2$ character projections commute with $X_b$; cyclic rotation makes the three nontrivial sectors unitarily equivalent. In the invariant sector, conjugation pairs the two complex cubic characters. The resulting lists $I, O, N$ satisfy

$$Z_n = I_n + 2O_n + 3N_n, \qquad Z_n(P) = I_n + 2O_n - N_n, \qquad Z_n(C) = I_n - O_n, \tag{2.2}$$

$$J_n = (Z_n + 3Z_n(P))/4, \qquad N_n = (Z_n - Z_n(P))/4,$$
$$I_n = (Z_n + 3Z_n(P) + 8Z_n(C))/12, \qquad O_n = (Z_n + 3Z_n(P) - 4Z_n(C))/12.$$

Each list is real and absolutely summable at the required moment orders. Even moments are nonnegative; odd ones can have either sign. No physical spin sector has been discarded by these spatial-sector identities. Repetition counts of two and three are essential in the source's dominant-weight argument.

At $b = 21/2$, the source uses $n = 6, 8, 10$ with all three twists and absolute error $10^{-5}$. Its filters are $x^6(-2 + 10x^2)^2$ for $I$ and $x^6(-25 + 100x^2)^2$ for $N$. They give the caps 100245/100000 and 88643/100000 after certified exterior-ray comparisons. At $b = 49/4$, the eighteen values $4 \le n \le 12$, $g = I, P$, have stated error $30 \cdot 10^{-6}$. The filters are

$$F_J(x) = x^4(7 + 17x - 280x^2 - 185x^3 + 1000x^4)^2, \qquad F_N(x) = x^4(12 - 394x^2 + 1000x^4)^2. \tag{2.3}$$

Their endpoint comparisons yield $|\lambda| < 100139/100000$ in $J$ and $|\lambda| < 872/1000$ in $N$. Positivity of the filters and control of entire exterior rays are necessary; a pointwise numerical grid would not suffice.

### 2.3 Thermal and variational finite premises

The two thermal appendices use the actual finite physical Hamiltonian, not an arbitrary fitted spectrum. Their contraction is

$$W = I - \frac{H_n(g) + (3n/2)I}{16}, \quad E = e^{\mu(W-I)}, \quad x_n = bn(3/2 - a), \quad Z_n(b, g) = e^{x_n} \|E\|_F^2, \tag{2.4}$$

where $\mu = 8b$. For $b = 21/2$, $\mu = 84$, degree $J = 256$, scale $Q = 2^{56}$, and $B = 16W$. For $b = 49/4$, $\mu = 98$, $J = 280$, $Q = 2^{55}$, and the integer matrix is $D = 32W$. The spectral interval from (1.3) is $[1 - 5n/32, 1]$; it lies in $[-1, 1]$ for all computed lengths. The exponential is replaced by a Poisson-conditioned polynomial, with error $< 2^{-116}$ in the first appendix and $< 2^{-84} < Q^{-1}$ in the second. Coefficient flooring, vector flooring, all omitted zero-integer-coefficient steps, orbit multiplicities, and scalar-exponential tails have separate error charges.

The first appendix computes Eisenstein coordinates for the cyclic twist with arbitrary-precision packed integers, justifying its lane floor using a bias, a mask, and a bound below $2^{63}$. The second uses signed 64-bit multiplication with a row absolute-sum bound 76 and intermediate coordinates below $2Q$, and arbitrary precision for final squares. The proof of no overflow is inductive; it must precede reliance on exact machine arithmetic. Their signed or antiunitary rotation/reversal symmetries preserve *exact* column norms, not rounded-column covariance. Section 5 preserves that distinction.

In `trials.tex`, a fixed integer triple $A_s \in M_{26}(\mathbb{R})$ defines

$$\psi_N(s_0, \dots, s_{N-1}) = \operatorname{Tr}(A_{s_0} \cdots A_{s_{N-1}}), \quad T_A = \sum_s A_s \otimes A_s, \quad V_N = \operatorname{Tr} T_A^N. \tag{2.5}$$

For $B_{st} = A_s A_t$, its insertion matrix is

$$O_A = \sum_{s,t=-1}^{1} stB_{st} \otimes B_{st} + \sum_{s=-1,0} \sum_{t=0,1} (B_{st} \otimes B_{s+1,t-1} + B_{s+1,t-1} \otimes B_{st}).$$

Expansion of tensor traces and cyclic translation give $\|\psi_N\|^2 = V_N$ and $\langle \psi_N, H_N \psi_N \rangle = NU_N$, where $U_N = \mathrm{Tr}(T_A^{N-2} O_A)$. Neither $T_A$ nor $O_A$ is assumed positive or self-adjoint. Virtual weight-difference blocks have dimensions $1, 8, 32, 80, 136, 162, 136, 80, 32, 8, 1$, summing to 676. The asserted exact comparisons at $N = 72, 120$ imply

$$V_N > 0, \quad U_N < -aV_N, \quad E_0(N) < -Na, \quad Z_N(b) > 1 \quad \text{for every } b > 0. \tag{2.6}$$

The variational vectors occupy total weight zero, but bound the minimum of the full physical spectrum. A variational upper bound on ground energy alone is not a gap lower bound.

### 2.4 Initialization and assembly

For a capped nonnegative mass list $t_i \le h$, $\sum_i t_i \le m$, the source uses the convex extremal bound

$$\sum_i t_i^r \le \mathcal{C}(m, h, r) = \lfloor m/h \rfloor h^r + (m - h\lfloor m/h \rfloor)^r \quad (r \ge 1).$$

At $b = 21/2$, caps on moments of order ten produce bounds at orders thirty and 120. Raising physical Boltzmann factors to power $5/2$ reaches $b = 105/4$. The 120-site trial premise forces a unique spatial weight larger than 0.88; multiplicity puts it in $I$. A convex residual-mass polygon sharpens it through 0.99 to 0.998. This yields $S(60, 105/4) > 7/8$. A distinct argument with physical Boltzmann factors gives $T(120, 105/4) > 7/8$. These steps cannot interchange physical energies and spatial eigenvalues.

At $b_* = 49/4$, the source's twelfth-moment bounds are $m_J = 1.0657205$ and $m_N = 0.2783155$. Put $\ell = 999/1000$, $v_J = 100139/100000$, $v_N = 872/1000$, and

$$B(k) = 3\mathcal{C}(m_N, v_N^{12}, k/12), \quad R_A(k) = B(k) + (m_J - \ell^{12})^{k/12},$$
$$p_0 = 1 - \left(1 + R_A(36)/\ell^{36}\right)^{-2}, \quad q_0 = 1 - \left(v_J^{72} + R_A(72)\right)^{-2}. \tag{2.7}$$

The 72-site trial premise excludes the alternative in which every $J$ eigenvalue has modulus at most $\ell$. Removing a larger eigenvalue and bounding the remaining positive even moments proves that these are admissible spatial and physical defects at $(36, 49/4)$. Six coupled updates, rounded upward to multiples of $10^{-7}$, give the printed exact terminal pair

$$p_6 = \frac{84513}{10^7}, \qquad q_6 = \frac{27493}{5 \cdot 10^6}, \qquad (n_6, b_6) = (2304, 784). \tag{2.8}$$

Assembly applies one abstract bootstrap to each seed. It does not furnish an independent check of the thermal totals or trial contractions. The infinite-volume corollary is a local-observable subsequence statement; its limit does not establish equality of all subsequential limits.

### 2.5 Dependency status

The logical order is

spin normalization and energy bound $\longrightarrow$ physical thermal matrices and their errors,

bond histories and rotation sectors $\longrightarrow$ signed spatial moments and positive filters,

thermal totals plus trial contractions $\longrightarrow$ finite seeds $\longrightarrow$ coupled bootstrap and all even lengths.

Reading all nodes is not executing all finite certificates. The initial local collection omitted the advertised `verification/verify.py`, most calculation scripts, and the full trial JSON certificates. This was not evidence of absence from the public repository: both full trial certificates, matrix data, and the verifier were subsequently retrieved successfully from the exact pinned public paths into an author-owned audit directory. Thus the report distinguishes an exact downstream comparison *conditional on a recorded integer* from recomputing that integer from physical matrix entries. Parent review and executed checks are recorded in Section 7 and local sidecars.

# 3 Specific Ram correspondences and non-correspondences

## 3.1 Physical synthesis and Gram matrices

Private Volume 21, `ch01d-physical-metrics.tex`, definition `v21c01:physical-gram-definition`, starts with injective synthesis $J : \mathbb{C}^d \to H$, $G = J^*J > 0$, and $\|Jc\|^2 = c^*Gc$. Its proposition `v21c01:metric-symmetry` proves $GB = B^*G$ for the compression $B = G^{-1}J^*AJ$ of a symmetric operator. These are the precise representation principles needed for nonorthogonal numerical coordinates. For the orthonormal spin basis the physical Gram is the identity. For Eisenstein real coefficients it is not: Section 5 gives its actual 2-by-2 blocks.

The same file, `v21c01:residual-decomposition` and `v21c01:residual-identity`, states

$$\|(A-\lambda)Jc\|^2 = \|(B-\lambda)c\|_G^2 + \|(I-P)AJc\|^2\,, \qquad P = JG^{-1}J^*. \tag{3.1}$$

It prevents a small finite residual of a compressed $X_b$ from being treated as a spectral certificate for the full history space. The source avoids this issue for its thermal inputs by retaining the entire finite physical Hilbert space, with exact symmetry accounting. A future finite history-space approximation would need a bound on the omitted term, not only a positive Gram matrix. The file's `v21c01:gram-equivalence` additionally requires relative quadratic-form bounds to transport an approximate metric.

## 3.2 Positive moments without an unjustified finite measure

Private Volume 21, `ch02/finite.tex`, defines `v21c02:hankel` and `v21c02:localizer` for an explicitly positive finite measure, with absolute moment integrability and specified support. The natural unweighted counting measure of all nonzero eigenvalues of $X_b$ need not be finite. Section 6 therefore weights it by $\lambda^4$ (or $\lambda^6$ for the first initialization). That supplies the missing finite mass and exactly matches the available moment orders. The resulting Hankel identities are classical moment identities, instantiated in the private Ram certificate language.

The same chapter's `v21c02:zero-energy` and `v21c02:atomic-certificate` require an exact zero quadratic form, real roots, positive weights, and a full moment check to identify an atomic measure. Our thermal intervals do not give those conditions. A nearly singular numerical Hankel matrix is not a rank-one transfer proof. The report uses positive filter inequalities, not unjustified atomic recovery.

## 3.3 Polynomial actions, not a new exponential algorithm

Private `TR-core-pat-5.tex`, subsection "Polynomial evaluation without eigendecomposition," equation `eq:filtererror`, bounds a self-adjoint spectral multiplier error by a scalar uniform approximation error. This is precisely the route from the scalar Poisson-conditioned series to the operator

error in (2.4). It is classical functional calculus; the private representation contributes an explicit contract for spectral interval, norm, polynomial error, and arithmetic error.

The report's Schur residual proposition `prop:schurresidual` and Woodbury theorem `thm:woodbury` were also read. They are not automatically shortcuts here. Eliminating coordinates in an exponential requires a justified dynamical or resolvent construction; a Schur complement of $H$ does not have the same heat trace. A nonzero seam perturbation has the form $\Delta h \otimes I_{3^{n-2}}$, up to factor permutation, so its rank is $\operatorname{rank}(\Delta h)3^{n-2}$. It is local in sites, not uniformly low rank on the full physical space. Any useful sector-specific low-rank property must be proved, not assumed from spatial locality.

### 3.4 Why the differential Ram formula is not used blindly

Private Volume 1, revision 2, `ch03-local-moments-rev2.tex`, theorem `thm:ram-local`, expresses a sufficiently regular kernel action as a finite Taylor differential operator plus an integral remainder, with bound

$$|r_N(x)| \le \frac{B_{N+1}}{(N+1)!} \int \|u\|^{N+1} \, |k_x(u)| \, du.$$

The labels `eq:ram-integral-remainder` and `eq:ram-remainder-bound` state the required derivative and domination hypotheses. The history space here is a disjoint union of simplexes of unbounded dimension, with discrete labels and chronological product changes across collision strata. No global Euclidean smoothness or finite derivative remainder has been proved for it. Substituting a differential Ram operator for $X_b$ would therefore introduce, not solve, a missing premise.

The same chapter's `thm:degree-closure` distinguishes polynomial exactness from invariant finite-dimensional closure; `eq:ram-jet-product` includes all coefficient derivatives under differentiation. The source's finite polynomial in $W$ is a functional-calculus polynomial on a fixed finite space, not a Taylor localization of $k_b$. Keeping these two uses of "polynomial" separate is a substantive safeguard. No private passage is being credited with proving the source's transfer construction, long-chain bootstrap, or trial inequalities.

## 4 Sharp inversion of purity and a conditional gap refinement

For an even $n \ge 4$ and $b > 0$, define

$$S(n,b) = \frac{Z_{2n}(b)}{Z_n(b)^2}, \qquad T(n,b) = \frac{Z_n(2b)}{Z_n(b)^2}. \tag{4.1}$$

The spatial probabilities are $|\lambda_i(b)|^n / Z_n(b)$. The physical probabilities list *every* eigenvector, $w_k = e^{-b(E_k(n)+an)}/Z_n(b)$. Both purities lie in $(0,1]$; the scalar shift cancels. The assumption that the same $Z$ has both representations is indispensable.

### 4.1 An exact scalar inverse

**Theorem 4.1** (Sharp purity-to-mass inverse)**.** *Let $(w_i)$ be a finite or countable nonnegative probability list and $P = \sum_i w_i^2 > 1/2$. Then its maximum $m$ is attained, is unique, and*

$$m \ge \frac{1 + \sqrt{2P-1}}{2}. \tag{4.2}$$

*For every $P \in (1/2, 1]$ this bound is attained by a probability list supported on at most two points.*

*Proof.* The supremum $s$ of the weights is positive. Only finitely many weights can exceed $s/2$, and they contain every sequence tending to $s$, so the supremum is attained. Since $P \le m \sum_i w_i = m$, we have $m > 1/2$, and no two weights equal $m$. Group the remaining weights: their sum of squares is at most the square of their sum. Consequently $P \le m^2 + (1-m)^2$. The right side is strictly increasing on $[1/2, 1]$, and its inversion yields (4.2). Equality holds for weights $(m, 1-m)$ with the displayed $m$; at $P = 1$ use a single weight. Countable grouping is valid by nonnegativity. □

**Corollary 4.2** (Physical gap from a purity lower bound)**.** *Let $H$ be a nonscalar finite-dimensional self-adjoint Hamiltonian, $b > 0$, and suppose the purity of its full Gibbs distribution is at least $P_* > 1/2$. Its ground state is unique, and*

$$\gamma \ge \frac{\Phi(P_*)}{b}, \qquad \Phi(P) = \log \frac{1 + \sqrt{2P-1}}{1 - \sqrt{2P-1}} \quad (1/2 < P < 1). \tag{4.3}$$

*Proof.* Ground-state eigenvectors have maximal Gibbs weight; uniqueness follows from a maximal weight greater than one half. For an eigenvector at the next distinct energy, $e^{-b\gamma} = w_1/w_0 \le (1-w_0)/w_0$. Apply Theorem 4.1 and take logarithms. In a nonscalar finite-dimensional Gibbs distribution all weights are strictly positive and $P < 1$, so the formula has no singular endpoint. The bound is sharp for two-level Hamiltonians. A degenerate first excitation only strengthens the bound if its multiplicity is separately known; none is assumed here. □

This improves the source's passage $w_0 \ge P$ to a binary-exact estimate. It does not estimate a gap between arbitrary compressed eigenvalues, and it gives no positive bound when only $P_* \le 1/2$ is known.

### 4.2 The squaring inequality is already sharp

For normalized squares $w_i^2/P$, the source proves

$$1 - P' \le f(1-P), \qquad f(u) = \frac{u^2}{2(1-u)^2}. \tag{4.4}$$

Indeed $1 - P' = 2\sum_{i<j} w_i^2 w_j^2 / P^2$ and $\sum_{i<j}(w_i w_j)^2 \le (\sum_{i<j} w_i w_j)^2$. For two positive weights equality holds. Every $P \in [1/2, 1]$ is a binary purity, so no smaller universal function of $P$ alone can replace (4.4) on that range. The improvement here is the terminal purity inversion and avoiding avoidable rounding, not a claim to improve the source's sharp squaring function.

### 4.3 Asymmetric finite certificates and intermediate lengths

Cancellation of partition functions gives

$$S(n, 2b) = S(n,b)^2 \frac{T(2n,b)}{T(n,b)^2}, \qquad T(4n,b) = T(2n,b)^2 \frac{S(2n,2b)}{S(2n,b)^2}.$$

If $1 - S(n,b) \le p$ and $1 - T(2n,b) \le q$, admissible next defects are

$$p^+ \ge f(1 - (1-p)^2(1-q)), \qquad q^+ \ge f(1 - (1-q)^2(1-p^+)), \quad p^+, q^+ < 1. \tag{4.5}$$

The second inequality must use the chosen rounded $p^+$, not an unrounded value from a different computation. The pair advances to $(2n, 2b)$.

**Proposition 4.3** (Length-resolved asymmetric gap certificate)**.** *Assume the physical and spatial representations above. Let $n \geq 4$ be even and suppose the defects are bounded by $p, q \in [0, 1)$. For every even $n \leq L \leq 2n$, put*

$$P_L = [(1-q)(1-p)^2]^{L/(2n)}. \tag{4.6}$$

*Then $T(L, b) \geq P_L$. Whenever $P_L > 1/2$, $H_L$ has a unique ground state and $\gamma_L \geq \Phi(P_L)/b$.*

*Proof.* For a nonnegative list, $\sum_i a_i^q \leq (\sum_i a_i^p)^{q/p}$ for $q \geq p > 0$, whenever the latter sum is finite. Normalize the $p$th powers to probabilities to prove it. Apply this to spatial eigenvalue moduli at temperatures $b$ and $2b$: $Z_L(b) \leq Z_n(b)^{L/n}$ and $Z_L(2b) \geq Z_{2n}(2b)^{L/(2n)}$. Their quotient is at least $[T(2n, b)S(n, b)^2]^{L/(2n)}$. Now use the two lower bounds and Corollary 4.2. No interpolation of physical eigenvalues between different lengths is assumed. □

With $p = p_6, q = q_6$ in (2.8), this is an explicit conditional bound at every even $2304 \leq L \leq 4608$. It retains the smaller physical defect and the actual position of $L$ in its dyadic interval. Further applications of (4.5) need only scalar rational arithmetic, not new thermal matrices; they can be used to restart the uniform criterion below at later lengths. They do not automatically improve the bound at earlier lengths.

### 4.4 Uniform propagation with exact seed retention

Define the increasing function

$$G(u) = \frac{1}{2}\big((1-u)^{-1} + (1-u)^{-2} + (1-u)^{-3}\big)^2. \tag{4.7}$$

Then $f(1-(1-u)^3) = G(u)u^2$.

**Theorem 4.4** (Conditional uniform certificate)**.** *Suppose the two representations hold for every even length and every positive inverse temperature. Let $n_0$ be even, $b_0 > 0$, $0 < u_0 < 1/6$, and suppose*

$$S(n_0, b_0), T(2n_0, b_0) \geq 1 - u_0, \quad G(u_0) \leq C, \quad r = Cu_0 < 1, \quad C(1-3u_0) > 3.$$

*Then for all even $L \geq n_0$ the ground state is unique and*

$$\gamma_L > \frac{\log(1/r)}{b_0}. \tag{4.8}$$

*Moreover, if $n_j = 2^j n_0$, $b_j = 2^j b_0$, $u_j = C^{-1} r^{2^j}$, then on $n_j \leq L \leq 2n_j$ the stronger finite-scale statement*

$$\gamma_L \geq b_j^{-1} \Phi\big((1-u_j)^{3L/(2n_j)}\big) \tag{4.9}$$

*holds.*

*Proof.* The common defects $u_j$ satisfy $u_{j+1} = Cu_j^2 < u_j$. In the first update (4.5), monotonicity bounds the result by $G(u_j)u_j^2 \leq u_{j+1}$. Use this same upper bound in the second update; it too is bounded by $G(u_j)u_j^2$. This proves the dyadic purity claims by induction. Proposition 4.3 gives $T(L, b_j) \geq (1-u_j)^{3L/(2n_j)} \geq (1-u_j)^3 \geq 1 - 3u_j > 1/2$ and hence (4.9). For the uniform strict statement, it suffices to use the weaker mass bound $w_0 \geq T$:

$$e^{-b_j \gamma_L} \leq \frac{3u_j}{1-3u_j} \leq \frac{3}{C(1-3u_0)} r^{2^j} < r^{2^j}.$$

Take logarithms, divide by $2^j b_0$, and cover all lengths by the dyadic intervals. The proof restates the source's classical bootstrap with all hypotheses exposed; it is not credited as a new theorem of the private corpus. □

**Corollary 4.5** (Retained-terminal-defect improvement)**.** *Conditional on the source's transfer identity and finite premises yielding* (2.8)*, set*

$$u_* = p_6, \qquad C_* = G(u_*), \qquad r_* = u_* G(u_*). \tag{4.10}$$

*All these are rational. Every even* $L \geq 2304$ *has a unique ground state and*

$$\gamma_L > \delta_* := \frac{1}{784} \log \frac{1}{u_* G(u_*)} > \frac{\log 20}{784}. \tag{4.11}$$

*Proof.* The exact source fractions give $0 < q_6 < p_6 < 1/100$. For $0 \leq u \leq 1/100$, $9/2 \leq G(u) \leq 9/[2(1-3u)^2] \leq 45000/9409 < 5$; the upper bound follows from $1-(1-u)^3 \leq 3u$ and the factorization of $G$. Hence $r_* < 5/100 = 1/20$, $C_*(1-3u_*) > (9/2)(97/100) > 3$, and all hypotheses of Theorem 4.4 hold. The strict comparison of logarithms proves the improvement. This proves a valid certificate without needing to round $C_*$ to a decimal; any upward rational approximation may instead be used if all inequalities are retained. □

The gain is conditional on exactly the same source finite inputs; no extra many-body calculation is required. The sharp binary inverse improves finite-scale constants but, by itself, does not improve the exponential decay rate $\log(1/r)/b_0$ as $j \to \infty$. Separating these mechanisms prevents an overstated asymptotic conclusion.

### 4.5 Seven exactly certified restarts

The parent independently evaluated the following finite scheme using rational arithmetic. It retains the asymmetry for a finite number of steps before restarting Theorem 4.4. Define

$$\lceil x \rceil_d = 10^{-d} \lceil 10^d x \rceil, \qquad (\widehat{p}_0, \widehat{q}_0) = (p_6, q_6),$$

and for $k = 0, \ldots, 5$ set

$$\begin{aligned} \widehat{p}_{k+1} &= \left\lceil f\big(1-(1-\widehat{p}_k)^2(1-\widehat{q}_k)\big) \right\rceil_{300}, \\ \widehat{q}_{k+1} &= \left\lceil f\big(1-(1-\widehat{q}_k)^2(1-\widehat{p}_{k+1})\big) \right\rceil_{300}. \end{aligned} \tag{4.12}$$

For each row $0 \leq k \leq 6$, let

$$n_k = 2304\, 2^k, \quad b_k = 784\, 2^k, \quad u_k = \max(\widehat{p}_k, \widehat{q}_k), \quad C_k = \lceil G(u_k) \rceil_8, \quad r_k = C_k u_k. \tag{4.13}$$

All displayed decimal numbers in Table 1 are terminating rationals, not rounded empirical gap estimates. The exact high-precision defects and products $r_k$ are stored in the parent receipt; equations (2.8) and (4.12) uniquely specify them without a long decimal table.

**Theorem 4.6** (Certified eventual improvement)**.** *Assume the source's physical/spatial partition-function representation for all even lengths and positive temperatures, and the finite premises that imply* (2.8)*. Every row of Table 1 gives a unique ground state and* $\gamma_L > d_k$ *for all even* $L \geq n_k$*. In particular,*

$$\gamma_L > \frac{2363}{500000} \quad (L \geq 147456 \text{ even}), \qquad \Delta_1 \geq \frac{2363}{500000} = 0.004726. \tag{4.14}$$

| $k$ | all even $L \geq n_k$ | $b_k$ | $C_k$ | certified $d_k$ |
|---|---|---|---|---|
| 0 | 2304 | 784 | 4.65561621 | 0.004126 |
| 1 | 4608 | 1568 | 4.50465840 | 0.004308 |
| 2 | 9216 | 3136 | 4.50000305 | 0.004492 |
| 3 | 18432 | 6272 | 4.50000001 | 0.004613 |
| 4 | 36864 | 12544 | 4.50000001 | 0.004678 |
| 5 | 73728 | 25088 | 4.50000001 | 0.004710 |
| 6 | 147456 | 50176 | 4.50000001 | 0.004726 |

Table 1: Conditional uniform gap bounds $\gamma_L > d_k$ from the unchanged source premises. The threshold length grows with the restart; the last row does not apply at $L = 2304$.

*Proof.* Start with the exact source pair, not its one-percent relaxation. Repeated use of (4.5), with upward rounding at each of the two ordered operations, proves by induction that $(\widehat{p}_k, \widehat{q}_k)$ bounds the defects at $(n_k, b_k)$. Exact rational checks establish, for each of the seven rows,

$$0 < u_k < 1/6, \quad G(u_k) \leq C_k, \quad r_k < 1, \quad C_k(1 - 3u_k) > 3.$$

Theorem 4.4 therefore gives $\gamma_L > \log(1/r_k)/b_k$ at every even $L \geq n_k$.

For completeness, the certification of the logarithm uses no floating-point inequality. Given the rational target $d_k$ in the table, put $x = b_k d_k$ and $N = \max(100, 5\lfloor x \rfloor + 40)$. Then $0 < x < N + 2$ and

$$e^x \leq U_N(x) := \sum_{j=0}^{N} \frac{x^j}{j!} + \frac{x^{N+1}}{(N+1)!} \frac{1}{1 - x/(N+2)}. \tag{4.15}$$

The ratio of consecutive tail terms is at most $x/(N+2) < 1$, proving the enclosure by a geometric series. The independently executed rational comparisons give $r_k U_N(b_k d_k) < 1$ for each row. Hence $e^{b_k d_k} r_k < 1$, or $d_k < \log(1/r_k)/b_k$. These finitely many comparisons, their exact operands, and the checked bootstrap conditions are recorded in `M543-parent-check-results.json`, generated by the separately read parent script. A floating logarithm in that script only selected candidate $d_k$; the acceptance test is the exact inequality just proved. The final row and a limit inferior prove (4.14). □

This is a fully specified finite scalar refinement of the same analytic scheme, conditional on the inherited seeds and transfer theorem. It needs neither altered thermal data nor a newly computed spectrum. The improvement over the source's eventual constant is meaningful but limited: it gives no approach to a sharp physical gap, and the strongest displayed bound begins at a much larger ring length.

### 4.6 A less restrictive terminal prefactor

**Proposition 4.7** (Binary-prefactor variant)**.** *In Theorem 4.4, replace $u_0 < 1/6$ and $C(1 - 3u_0) > 3$ by*

$$(1 - u_0)^3 > \tfrac{1}{2}, \qquad C > \kappa(u_0) := \frac{6}{\left(1 + \sqrt{2(1-u_0)^3 - 1}\right)^2}. \tag{4.16}$$

*Keep $G(u_0) \leq C$ and $0 < Cu_0 < 1$. Then the same uniform strict bound (4.8) follows.*

*Proof.* The dyadic induction did not use 1/6 or the old prefactor condition. Write $P = T(L, b_j) \geq (1 - u_j)^3 > 1/2$. Corollary 4.2, before logarithms, gives

$$e^{-b_j \gamma_L} \leq \frac{1 - \sqrt{2P-1}}{1 + \sqrt{2P-1}} = \frac{2(1-P)}{(1+\sqrt{2P-1})^2} \leq \kappa(u_0) u_j.$$

Here $1 - P \leq 3u_j$ and the denominator is at least its value at $(1 - u_0)^3$. Therefore the ratio is at most $(\kappa(u_0)/C) r^{2^j} < r^{2^j}$. This proves the assertion with unchanged coverage of lengths. □

This is an extension of the sufficient certificate, not a newly verified family of Hamiltonians. Simultaneous feasibility with $G(u_0)u_0 < 1$ still restricts the seed. No claim is made that this variant further improves the selected $C_* = G(u_*)$.

# 5 Physical-Gram rounding and trace-error certificates

## 5.1 The actual Eisenstein metric

Let $\omega = e^{2\pi i/3}$. Writing $z = u + v\omega$ uses two *real* coordinates. The real-linear synthesis $J : \mathbb{R}^2 \to \mathbb{C}$ is injective; it would not be injective if incorrectly interpreted as a complex-linear map from $\mathbb{C}^2$. Its physical Gram matrix is

$$|u + v\omega|^2 = u^2 - uv + v^2 = \begin{pmatrix} u & v \end{pmatrix} G_\omega \begin{pmatrix} u \\ v \end{pmatrix}, \qquad G_\omega = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix}. \tag{5.1}$$

The eigenvalues are $1/2$ and $3/2$. For $d$ complex coordinates use the direct sum of $d$ such blocks. A physical contraction $W$ is a contraction in this real Gram norm after coordinate transport; its real coordinate matrix need not be a Euclidean contraction.

**Lemma 5.1** (Sharp coordinate-floor bound)**.** *For real $u, v$, define $\mathrm{fl}(u + v\omega) = \lfloor u \rfloor + \lfloor v \rfloor \omega$. Then*

$$|z - \mathrm{fl}(z)| \leq 1. \tag{5.2}$$

*For a complex vector of dimension $d$, coordinatewise Eisenstein flooring has physical Euclidean error at most $\sqrt{d}$. The constant one per coordinate is the least uniform upper bound.*

*Proof.* Let $x = u - \lfloor u \rfloor$ and $y = v - \lfloor v \rfloor$, so $0 \leq x, y < 1$. Its squared physical error is $F(x, y) = x^2 - xy + y^2$. For fixed $y$, convexity in $x$ bounds $F$ by its values at $x = 0, 1$, and each of those is a convex function of $y$ with maximum at $y = 0, 1$. The four vertex values are $0, 1, 1, 1$. Summing squared errors proves the vector bound. Taking $x \uparrow 1, y = 0$ shows that no smaller uniform constant works. The proof uses the signs and square constraint on the two floor remainders; a bare eigenvalue bound on $G_\omega$ would lose information. □

The source `thermal120.tex`, `eq:thermal120-column-error`, uses the valid but coarser $2\sqrt{d}$. The sharper $\sqrt{d}$ applies to the *same algorithm and same output integers*, including negative original coefficients, because floor remainders still lie in $[0, 1)$. It is not a new rounding rule and requires no claim about lucky cancellation. For real coordinates the source's second thermal appendix already uses $\sqrt{d}$.

### 5.2 A column and orbit-sensitive theorem

**Theorem 5.2** (Horner-to-trace physical error)**.** *Let $W$ be a self-adjoint contraction on $\mathbb{C}^D$ decomposed into orthogonal invariant blocks. Let $p_j \geq 0$, $\sum_{j=0}^{J} p_j = 1$, and $P(W) = \sum_{j=0}^{J} p_j W^j$ approximate an operator $E$ with $\|P(W) - E\| \leq \tau$. For a unit coordinate vector $e_r$ in a block of dimension $d_r$, let $X_{J+1} = 0$ and compute*

$$X_j = \mathrm{fl}(W X_{j+1}) + \lfloor Q p_j \rfloor e_r, \qquad j = J, J-1, \ldots, 0, \tag{5.3}$$

*in exact integer or rational arithmetic before flooring, with real or Eisenstein-coordinate flooring and $Q > 0$. Then*

$$\|X_0 - QEe_r\| \leq \eta_r := (J+1)(\sqrt{d_r} + 1) + Q\tau. \tag{5.4}$$

*Suppose representatives have positive integer multiplicities $\nu_r$ and their exact column norms reproduce the full Frobenius norm: $\sum_r \nu_r \|Ee_r\|^2 = \|E\|_F^2$. Let*

$$W_{\mathrm{int}} = \sum_r \nu_r \|X_{0,r}\|^2, \qquad \mathcal{E} = \left( \sum_r \nu_r \eta_r^2 \right)^{1/2}. \tag{5.5}$$

*Then $|\sqrt{W_{\mathrm{int}}} - Q\|E\|_F| \leq \mathcal{E}$. If $Z = e^x \|E\|_F^2$ and $s_- \leq e^x \leq s_+$ with $s_- \geq 0$, certified endpoints are*

$$\frac{s_-}{Q^2} \max(0, \sqrt{W_{\mathrm{int}}} - \mathcal{E})^2 \leq Z \leq \frac{s_+}{Q^2} (\sqrt{W_{\mathrm{int}}} + \mathcal{E})^2. \tag{5.6}$$

*Proof.* The ideal suffix $Y_j = Q \sum_{\ell=j}^{J} p_\ell W^{\ell-j} e_r$ has $Y_j = W Y_{j+1} + Q p_j e_r$. Subtraction and the physical contraction property give $\|X_j - Y_j\| \leq \|X_{j+1} - Y_{j+1}\| + \sqrt{d_r} + 1$ by Lemma 5.1 (or the real-coordinate bound). Induction and the polynomial error prove (5.4). Form two abstract direct sums, repeating each computed column and its exact counterpart $\nu_r$ times. Their squared error is at most $\sum_r \nu_r \eta_r^2$. The reverse triangle inequality gives the stated norm comparison. Their exact norms are $\sqrt{W_{\mathrm{int}}}$ and $Q\|E\|_F$, regardless of the behavior of rounded columns under symmetries. Nonnegative squaring and the scalar enclosure finish the proof. □

For the source, $W_{\mathrm{int}}$ is an integer because (5.1) has integer values on integer coefficients. To obtain purely rational endpoints replace $\sqrt{W_{\mathrm{int}}}$ below by $r = \lfloor \sqrt{W_{\mathrm{int}}} \rfloor$ and above by $r + 1$, and replace $\mathcal{E}$ by any certified rational upper bound. The nonnegative truncation in the lower endpoint is essential. Exact block dimensions or orbit-specific errors can replace the source's common $3^n$ bound without requiring equivariance of the rounded recurrence.

**Corollary 5.3** (A revised first-appendix error budget)**.** *For every case $n = 6, 8, 10$, $g = I, P, C$ in the first thermal appendix, the same computed columns satisfy the rounding-only component of the column-error bound with*

$$257(\sqrt{3^n} + 1) \leq 257(243 + 1) = 62708 \tag{5.7}$$

*instead of* 125159*. The full column error additionally includes $Q\tau$ for the exponential approximation. Consequently the scaled norm error, including both components, is at most*

$$3^6\, 243 \left( \frac{62708}{2^{56}} + 2^{-116} \right) < 2 \cdot 10^{-7}. \tag{5.8}$$

*Conditional on the source's rational enclosure of its printed totals within $2 \cdot 10^{-6}$ of each center, the final trace error is therefore less than*

$$2 \cdot 10^{-6} + (8 + \epsilon_*)\epsilon_* < \frac{9}{2500000}, \qquad \epsilon_* = 3^6\, 243 \left( \frac{62708}{2^{56}} + 2^{-116} \right). \tag{5.9}$$

*Proof.* Use Theorem 5.2, the source's $\tau < 2^{-116}$, $e^{x_n/2} < 3^6$, and $\sqrt{3^n} \le 243$. The rational strict comparison in (5.8) is checked directly. The source's total enclosure bounds the computed scaled norm by 4. If its distance to the true scaled norm is at most $\epsilon_*$, the difference of squares is at most $(8+\epsilon_*)\epsilon_*$. The parent's exact rational comparison, retaining $\epsilon_*$ rather than replacing it by $2 \cdot 10^{-7}$ at this final step, gives (5.9). □

This certifies a tighter analytic error allowance for unchanged integer totals. It does not by itself recompute those totals, improve the first initialization with newly optimized filters, or show a faster computation. The lower error budget can be propagated in future interval arithmetic, but no unexecuted propagation is credited as a realized numerical gain.

### 5.3 A posteriori residual refinement

The column proof admits an optional stronger certificate. If the exact local recurrence defect is $r_j = X_j - WX_{j+1} - Qp_j e_r$, then

$$X_0 - QP(W)e_r = \sum_{j=0}^{J} W^j r_j, \qquad \|X_0 - QEe_r\| \le \sum_{j=0}^{J} \|r_j\| + Q\tau. \tag{5.10}$$

To prove the identity, unroll $X_j - Y_j = W(X_{j+1} - Y_{j+1}) + r_j$ from $j = 0$ to $J$, then use contraction. For Eisenstein rational coordinates, each squared residual norm is exactly a rational Gram form. This is a reproducible local refinement, not an assertion that residuals were logged in the source. It requires recording the actual coefficient fractions and floor remainders, including omitted steps; otherwise use the a priori theorem.

## 6 Positive weighted-moment certificates

### 6.1 A finite measure matched to the source data

**Proposition 6.1** (Weighted spectral moment representation)**.** *Let $X$ be compact self-adjoint with $\sum_i |\lambda_i|^4 < \infty$. Its nonzero eigenvalues, with multiplicity, define the finite positive measure*

$$\mu_4 = \sum_i \lambda_i^4 \delta_{\lambda_i}, \quad m_k = \int x^k \, d\mu_4 = \sum_i \lambda_i^{k+4} \quad (k \ge 0). \tag{6.1}$$

*For every $d \ge 0$ the Hankel matrix $H_d = (m_{i+j})_{i,j=0}^{d}$ is positive semidefinite. For every real polynomial $g \ge 0$ on the support, its localizing matrix $(\int g(x) x^{i+j} \, d\mu_4)$ is positive semidefinite. The same assertions hold on every reducing sector and with any even weight exponent $2k \ge 4$.*

*Proof.* Compactness bounds $|\lambda_i|$ and fourth-power summability gives finite mass and every higher absolute moment. For $c \in \mathbb{C}^{d+1}$, $c^* H_d c = \sum_i \lambda_i^4 |\sum_j c_j \lambda_i^j|^2 \ge 0$. Multiplying each summand by a nonnegative $g(\lambda_i)$ proves the localizer statement. Reducing sectors select sublists; replacing the fourth power by a higher even power preserves finiteness and positivity. □

The measure is not necessarily normalized; its mass is $M_4$, not one. Normalization by $M_4$ is allowed only if it is positive and any interval division is controlled. The source's $J$ and $N$ filters (2.3) are exactly integrals of polynomial squares with respect to their respective $\mu_4$. Their moment matrices have order 4 and use only powers 4 through 12. The first initialization uses $\mu_6$ and squares of a degree-one polynomial in $x^2$: the factor being squared is degree two in $x$, so its square contains $1, x^2, x^4$, yielding moments $6, 8, 10$. Zero eigenvalues contribute nothing and are harmless for an exterior spectral-modulus cap, but this weighting would not count their multiplicity.

### 6.2 A robust certificate with explicit thermal errors

**Theorem 6.2** (Interval polynomial-square certificate)**.** *Let a real eigenvalue list satisfy the assumptions of Proposition 6.1. Suppose $|M_j - c_j| \le \epsilon_j$ for all required $j \ge 4$. Choose a real positive semidefinite rational matrix $A = (A_{kl})_{k,l=0}^{d}$, let $v(x) = (1, x, \dots, x^d)^T$, and write*

$$F(x) = x^4 v(x)^T A v(x) = \sum_{j=4}^{2d+4} f_j x^j, \qquad B = \sum_j f_j c_j + \sum_j |f_j| \epsilon_j. \tag{6.2}$$

*Then $0 \le \sum_i F(\lambda_i) \le B$. If $F(x) > B$ for all real $|x| \ge v_* > 0$, every $|\lambda_i| < v_*$. More generally, if $F \ge a_* > 0$ on a set $K$, then the number of eigenvalues in $K$, with multiplicity, is at most $B/a_*$. If $q(x)^2 \ge a_*$ on $K$ and $F(x) = x^4 q(x)^2$, then $\mu_4(K) \le B/a_*$.*

*Proof.* Positive semidefiniteness gives $F(x) \ge 0$ on the real line. Absolute convergence permits summing the finite polynomial expansion and applying each interval error to obtain $\sum_i F(\lambda_i) \le B$. One eigenvalue in the exterior with $F > B$ contradicts this bound. Summing the lower bound $a_*$ over eigenvalues in $K$ proves the count bound. For the weighted-mass statement use $\sum_{\lambda_i \in K} \lambda_i^4 q(\lambda_i)^2 \ge a_* \mu_4(K)$. □

The rank-one choice $A = qq^T$ reproduces the source's square filters exactly. A sum of squares allows more flexible certificates, but no optimization success is claimed here. A computer-produced $A$ must have exact or validated positive semidefiniteness; a floating-point solver status is not sufficient. Polynomial positivity on an exterior ray requires a proof, such as the source's nonnegative translated coefficients, or a certified root/Sturm analysis.

When the moments are linear functions $M = Lz$ of a vector of thermal observations with centers $c$ and independent component bounds $\epsilon_a$, error should be propagated before discarding correlations:

$$f^T M \le f^T L c + \sum_a |(L^T f)_a| \epsilon_a. \tag{6.3}$$

This follows by maximizing a linear form over its specified rectangular uncertainty set. A known smaller joint uncertainty set can sharpen this support bound, but cannot be assumed without evidence. Equation (6.3) is especially useful for combined sector filters: treating intermediate sector intervals as independent may lose cancellations from shared $Z_n$. For a single $J$ or $N$ square filter with independent errors at each order, no automatic numerical gain follows.

### 6.3 What this reformulation gains and does not gain

The weighted measure makes the positivity assumptions and available degree budget explicit on the actual infinite spatial list. It also supplies a precise way to test or design additional filters with a finite rational witness. It does not recover all eigenvalues from finitely many moments, certify the eigenvectors, or identify a finite-rank $X_b$. The source already uses valid positive polynomial sums without naming this measure. Accordingly, we credit a clarified and extensible certificate representation, not a correction of an established flaw in the source or a novel moment-problem theorem.

## 7 Verification, review, and reproducibility

### 7.1 Executed scope and division of work

All ten mathematical source files were read in full. The source bibliography, both local README files, and selected output records were inspected. The two Volume 21 chapters cited here were

read in full; the specified local-moment and core-patent passages were read in their actual files, not inferred from a routing map. Exact paths and reading scope are listed in Appendix A; hashes are recorded in the author input manifest.

The parent's independent scalar receipt reports 62 exact checks passed: all eighteen second-appendix printed totals imply the stated trace intervals with margins exceeding $13 \cdot 10^{-6}$; all nine first-appendix totals imply their scalar enclosures; both polynomial-filter bounds and endpoint margins, the initial rational defects, and all six source rounded updates check exactly; all seven restart hypotheses and rational logarithm lower bounds pass; and the refined first-appendix error is below $9/2500000$. These checks use printed thermal totals and do not regenerate all 27 totals.

The parent separately regenerated four physical integer totals, at $n = 4, 5$ with twists $I, P$, matching all four printed values exactly. Full physical dimensions were $81,243$, with $13,22$ representative columns respectively. It used 281 Horner steps, signed 64-bit operations bounded by $76 \cdot 2Q < 2^{63} - 1$, and arbitrary-precision final squares and sums. This is not a replay at lengths 6 through 12 or at temperature $21/2$. The author read the scripts and receipts; these are independently executed parent results, not duplicate author executions.

The author's nine additional exact checks cover a 10201-point rational floor-cell regression, Gram invariance under multiplication by $\omega$, a signed-contraction Horner residual identity, fourth-weighted moment and Hankel identities with negative odd moment, and the revised error budget. Universal conclusions rely on the report's proofs, not on these samples. In a separate public-certificate check, both full recorded trial pairs $(U_N, V_N)$ were parsed at $N = 72,120$: $V_N > 0$, the displayed $10^{-12}$ scaled intervals, and the recorded positive margins $-700741V_N - 500000U_N$ all matched exactly. At $N = 120$ the recorded block numerators and norms sum to the recorded totals and their dimensions sum to 676. This verifies consequences and consistency of the full integers, not their generation by matrix powers.

The author also regenerated all 2028 scalar entries of the three 26-by-26 trial matrices directly from the source's entry rules, matching the downloaded matrix data exactly. Its file hash matches the matrix binding in the 120-site certificate. This small construction check does not evaluate the large matrix powers defining $U_N, V_N$.

The substantial source was made available to the parent before final formatting. The parent read the entire report and all local proofs, agreed with their conditional mathematical scope, and requested corrections to the transfer and limit-inferior notation, the first-filter polynomial-degree description, and the distinction between rounding-only and full column error. Those findings were incorporated before the second formatting round. The final manifest binds the source, report PDF, executed-check outputs, and input hashes. No delivery event is recorded by this author.

### 7.2 Reproducibility levels

There are four distinct levels: (1) an analytic derivation from explicit hypotheses; (2) exact rational comparisons using recorded finite integers; (3) a fresh matrix/column or contraction replay producing those integers; and (4) a full independent proof audit plus all required finite replays. The present work proves its local results at level (1), records all 27 downstream thermal comparisons and trial-integer comparisons at level (2), records only the four physical cases above at level (3), and does not claim level (4). The public verifier was downloaded for provenance but not executed; the complete dependencies for a full replay were not collected. Source hashes and printed outputs are not substitutes for unexecuted calculations.

All finite tests have explicit finite sizes and no unbounded search. Rational arithmetic is used for inequality predicates. Decimal logarithms, if reported, are display approximations, not proof premises. Source hashes bind bytes, not truth; a pin in the catalogue and a local hash inventory do

not alone authenticate a complete Git checkout. The local source tree has no Git metadata, so the pin is recorded as the catalogue's reference identity.

# 8 Comparison, scope, and future research

## 8.1 Public primary literature

A public literature check by mathematical topic and title was made on October 7, 2026; no private corpus text was submitted in queries. Haldane's original prediction concerns the difference between integer and half-odd-integer chains [2]. AKLT proves a gapped spin-one model with a biquadratic interaction, not the pure bilinear Hamiltonian (1.1) [3]. Yarotsky's perturbative stability result includes a neighborhood of the AKLT model, without supplying a path to this bilinear point [4]. Gosset–Mozgunov's sharper finite-size criterion applies to frustration-free systems, an assumption not supplied merely by shifting the energy of the Heisenberg chain [5].

Tasaki's 2025 result assumes a uniform gap for odd open chains with endpoint magnetic fields to infer a nontrivial topological index [6]. Its boundary hypothesis cannot be inferred simply by renaming an even-periodic estimate. The recent 2026 string-order bootstrap of Banerjee, Guo, and Chowdhury certifies a different observable using energy-window information [7]; it does not remove this report's finite thermal or variational premises. These are comparisons of hypotheses and target quantities, not a claim to have surveyed every subsequent paper.

The source's periodic matrix-product trial representation belongs to the established MPS framework [8]. Conditioning Poisson coefficients changes their total-variation mass by at most twice the omitted probability; this is the classical Fox–Glynn principle [9]. Self-adjoint polynomial functional calculus, Gram-coordinate transport, convex mass concentration, and moment-square certificates are likewise classical. The private Ram contribution is a disciplined representation and error-accounting framework instantiated here, not ownership of those general techniques.

## 8.2 Advantages and costs

Exact seed retention is cheap and changes a stated uniform constant, but depends on the same source seed proof. The binary purity inverse is sharp without dimension information and improves finite-length statements, but cannot recover missing information when purity is at most one half. The Gram-floor theorem improves an actual arithmetic error allowance without changing the calculation, but it neither validates the output integers nor demonstrates a speedup. Weighted moments avoid an infinite counting-measure ambiguity and allow rational certificate design, but noisy moments do not identify the transfer operator. Block-sensitive or a posteriori error bounds can be tighter at the cost of more metadata and bookkeeping.

## 8.3 Next research steps

First collect the remaining pinned verifier dependencies and perform a resource-bounded full replay, including the unrecomputed large trial contractions, in an explicitly owned output directory. Second, propagate the revised first-appendix error budget through optimized interval filters and quantify whether it changes the 60-site initialization; the current report does not claim that unexecuted gain. Third, extend the bounded asymmetric restart search beyond the seven certified rows only with an explicit stopping budget, measuring the resulting eventual gap and threshold length separately. Fourth, search for positive-semidefinite rational polynomial filters using the same moment budget, verifying every candidate independently. Fifth, investigate a finite history-space representation only

after deriving its physical Gram matrix and an omitted-space operator bound. Any extension to boundary fields, thermodynamic uniqueness, or topological order must supply those new hypotheses explicitly.

### 8.4 Contribution assessment

The realized local content consists of the sharp terminal purity inverse and length-resolved certificate, an exact retained-seed conditional uniform improvement, a binary-prefactor extension, the Eisenstein and orbit-sensitive trace-error theorems, and a finite weighted-moment formulation. The source's transfer operator, source trial vectors, inherited bootstrap mechanism, and unreplayed thermal totals are not counted as new contributions. These supported local refinements justify C3 rather than a score driven by L5 applicability or the headline theorem's importance. A higher rating would require evidence of a more consequential realized advance, such as a substantial new certified regime or closure of an important unmet dependency, not merely a numerical factor or report length.

## A Exact local paths and reading scope

The common report root is `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/`. Under the pinned source directory specified in Section 1, full mathematical reads covered `build/main.tex`, `introduction.tex`, `model.tex`, `transfer.tex`, `bootstrap.tex`, `initialization.tex`, `assembly.tex`, `thermal120.tex`, `thermal72.tex`, and `trials.tex`. This includes the two thermal appendices, the full variational appendix, and the infinite-volume consequence, not only the catalogue screen. The bibliography was read for attribution, and the source and computation README files were read for the claimed verification workflow.

Actual private readings were:

1. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex`, entire file. Sections "A Nonorthogonal Representation Carries a Metric," "The Full Residual Has an Omitted Component," "Approximating the Metric," and "Finite Measurements Do Not Give a Continuum Frame." Exact labels used are listed in Section 3.

2. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch02/finite.tex`, entire file. Sections "What a Moment Record Specifies," "Finite Atomic Recovery With a Complete Check," and "Finite Nullspaces and Identified Observables."

3. `C:/Users/mural/OneDrive/Documents/RamPatsCore50-TRs/TR-core-pat-5/TR-core-pat-5.tex`, lines 288–464: sampled/discarded errors; "Polynomial evaluation without eigendecomposition"; "Coarsening, Lifts, and Schur Sources"; and low-rank identities and residual/conditioning qualifications in "Topology Edits and Checked Low-Rank Reuse." No claim of an entire-report read is made.

4. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume1-rev2/chapters/ch03-local-moments-rev2.tex`, lines 1–260: output and source localizations, the two normalizations, polynomial exactness versus closure, and the differentiated local formula. Referenced auxiliary input files were not treated as read; the formulas used here are in the indicated parent file.

The contribution policy at `contribution-policy.json` and assigned row of `00-Applicability-Catalogue/applicability.json` were read before authoring. The PDF skill and its required

creation marker were read/executed before the first authoring command. These readings and their SHA-256 digests are recorded in `_audit/M543-author-inputs.json`.

## B Independent local verification specifications

The author script `_audit/M543-author-checks.py` contains finite exact tests for the new local algebra. Its input assertions and output record distinguish sample verification from proofs:

1. Verify the Gram-square floor inequality on all rational pairs in a fixed 101-by-101 mesh of the unit square, and the four exact vertex maxima. The mesh is a regression test only; separate convexity proves the full square.

2. Check the revised 62708 column bound, scaled error below $2 \cdot 10^{-7}$, and trace error below $4 \cdot 10^{-6}$ as exact rational inequalities. No printed thermal total is regenerated by this test.

3. Check exact real-Gram norm preservation under multiplication by $\omega$ and the a posteriori Horner residual telescoping identity for a rational negative contraction. The parent separately checked the binary algebra; the report supplies the full sharpness proof.

4. Construct an explicit signed real eigenvalue list, its fourth-weighted moments and Hankel quadratic forms using rational arithmetic. Verify its square-filter sum both by eigenvalue summation and by moment coefficients, including a negative odd moment.

The parent conducts a different set of source-specific rational and bounded physical checks. The report preserves that division to avoid inflating evidence by counting a single replay twice.

## References


[1] OpenAI, *The periodic spin-one Haldane gap*, September 24, 2026, source pin `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; local mathematical files listed above. https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/The-periodic-spin-one-Haldane-gap-September-24-2026/paper.pdf.

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[10] M. Subbarao, confidential Ram Calculus corpus, Volume 21, *A Nonorthogonal Representation Carries a Metric* and subsequent sections, exact local path and labels in Section 3 and Appendix A.

[11] M. Subbarao, confidential Ram Calculus corpus, Volume 21, finite positive-moment chapter, exact local path and labels in Section 3 and Appendix A.

[12] M. Subbarao, confidential *TR-core-pat-5*, polynomial-filter, Schur, and low-rank-update passages, exact local path and labels above.

[13] M. Subbarao, confidential Ram Calculus corpus, Volume 1, revision 2, *Local Moments and Ram Differential Operators*, exact local path and labels above.